A MODERN LOOK AT CLASSICAL GENERALIZATION BOUNDS

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Abstract.

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1. Introduction

In machine learning, there exists a large number of algorithms, notably neural networks trained with gradient descent, which achieve great performances at prediction tasks on unseen data. A big question in learning theory is explaining what drives such *generalization* [5, 4]. Generalization is the process of applying a model to unseen data, after the algorithm has learned it from some available data. A number of results have tried to answer this question, coming to the apparent conclusion that the classical theory is not able to exhaustively explain what drives the generalization ability of learning algorithms. In particular, it fails to describe the behavior of the test error of learning algorithms as the complexity of the hypotheses set increases [1, 5, 6, 3].

This thesis seeks to understand whether this is true by evaluating qualitatively theoretical results against empirical evidences, in the setting of a finite hypotheses set of binary decision trees. We look for a bound on the *generalization gap* as the maximum depth of decision trees increases. The generalization gap is defined as the expected difference between the expected and empirical risk of the selected model,

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right]$$

where the expectation is taken over the disorder of the data.

In particular, we consider a number of existing theoretical results that increasingly account for the structure of the learning problem at hand, namely the finiteness of the data sample and the noise in the data. For each one of these results, we inquire its descriptive power by carrying out experiments and comparing it with the empirical generalization gap in asymptotic terms. Our aim is to empirically test the bounds as functions that capture the qualitative behavior of the gap curve. This means that we are not concerned with finding a quantitatively exact bound on the generalization gap, but one that reproduces its shape. Indeed, the bounds we consider are distirbution-free and use an elementary technology.

By doing so, it is possible to examine how the quantities appearing in the bounds affect their shape, and thereby identify which aspects of the learning problem influence the generalization ability of decision trees.

The final result of this thesis is a bound on the generalization gap of decision trees that is empirically validated as a function describing the shape of the gap, providing a first step to the theoretical understanding of the behavior of decision trees on unseen data. It also develops a framework and a toolkit that can be extended to the study of the generalization ability of other learning algorithms, like gradient-based ones.

1.1. **Setting.**

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Appendix A. Proofs

Proof of Theorem **??**. Let $Z \coloneqq (X, Y)$ and $Z_i \coloneqq (X_i, Y_i)$, so that

$$R(h) = \mathbb{E}[\ell_h(Z)] \qquad \text{and} \qquad \hat{R}_D(h) = \frac{1}{n}\sum_{i=1}^{n} \ell_h(Z_i).$$

Introduce an independent copy $D' = (Z'_1, \ldots, Z'_n)$ of the dataset D. Since the Z'_i are distributed as Z and are independent of D, we rewrite

$$R(h) = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i) \,\middle|\, D\right].$$

Then,

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] = \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right) \,\middle|\, D\right]\right].$$

since $\ell_h(Z_i)$ is a function of D, hence $\mathbb{E}\left[\ell_h(Z_i) \,|\, D\right] = \ell_h(Z_i)$. Since the supremum of expectations is at most the expectation of the supremum, and by the law of total expectation, we obtain

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] \leq \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right)\right]. \tag{1}$$

Now we apply the symmetrization trick. Let $\mathscr{H} = \{h_1, \ldots, h_m\}$, with $m = |\mathscr{H}|$. Since Z_i and Z'_i are i.i.d. for each $i = 1, \ldots, n$, swapping them does not change the joint distribution, i.e.

$$\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \overset{d}{=} \left(\ell_{h_j}(Z_i) - \ell_{h_j}(Z'_i)\right)_{j=1}^{m}.$$

This means that, for any fixed $(e_1, \ldots, e_n) \in \{-1, 1\}^n$,

$$\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \overset{d}{=} \left(e_i\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

The pairs $(Z_1, Z'_1), \ldots, (Z_n, Z'_n)$ are also independent, therefore equality in distribution holds also when taking the sum, i.e.

$$\left(\frac{1}{n}\sum_{i=1}^{n} \left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m} \overset{d}{=} \left(\frac{1}{n}\sum_{i=1}^{n} e_i\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

Finally, because the m random variables are jointly equally distributed, we have equality in distribution of the suprema

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right) \overset{d}{=} \sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} e_i\left(\ell_h(Z'_i) - \ell_h(Z_i)\right).$$

Because the two sides in the previous equation have the same distribution, their expected value is the same

$$\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right] = \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}e_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right].$$

Note that until now we have considered only the disorder of the data D, D'. Because this holds for any value of $(e_1,\dots,e_n)\in\{-1,1\}^n$, it holds surely for the random variable $\xi=(\xi_1,\dots,\xi_n)$, i.e.

$$\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right] = \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\,\middle|\,\xi\right]$$

surely. Taking the expected value over the disorder of ξ, we obtain

$$\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right] = \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right]$$

where the left hand side is independent of ξ and in the right hand side we use the law of total expectation. Plugging the result into (1), we have

$$\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}(R-\hat{R}_D)(h)\right] \le \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right]$$

Now we split the sum over $i=1,\dots,n$ into

$$\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right) = \frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i') + \frac{1}{n}\sum_{i=1}^{n}\left(-\xi_i\ell_h(Z_i)\right)$$

Then, we use that the supremum of the sum is bounded by the sum of the suprema,

$$\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')+\frac{1}{n}\sum_{i=1}^{n}\left(-\xi_i\ell_h(Z_i)\right) \le \sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')+\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(-\xi_i\ell_h(Z_i)\right)$$

and that the expectation of the sum is the sum of the expectations, obtaining

$$\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}(R-\hat{R}_D)(h)\right] \le \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')\right] + \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}(-\xi_i)\ell_h(Z_i)\right]$$

We also observe that we can replace $-\xi_i$ with ξ_i, because they have the same distributions, hence the same expected value, and are independent of the datasets. Therefore,

$$\begin{aligned}\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}(R-\hat{R}_D)(h)\right] &\le \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')\right] + \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i)\right]\\ &= 2\mathcal{R}_n(\mathscr{L}).\end{aligned}$$

□

Proof of Lemma **??**. For any $t > 0$,

$$\begin{aligned}
\exp\left(t\,\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) &\le \mathbb{E}\left[\exp\left(t\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&= \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&\le \mathbb{E}\left[\sum_{\ell_h\in\mathscr{L}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&= \mathbb{E}\left[\sum_{\ell_h\in\mathscr{L}}\prod_{i=1}^{n}\mathbb{E}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\,\middle|\,D\right]\right] \quad (2)
\end{aligned}$$

where we use Jensen's inequality for the convex function $\exp(\cdot)$ in the first inequality, that exp is increasing in the first equality, we bound the supremum with the sum (all terms are positive) in the second inequality, and we use the independence of the ξ_i's, and their independence of D, in the last equality.

Now we apply Hoeffding's lemma (Theorem **??**) to each factor for fixed $(X_i, Y_i) = (x_i, y_i)$ taking any value in $\mathcal{X} \times \mathcal{Y}$. The random variable $\xi_i\ell_h(x_i, y_i)$ is centered, since $\mathbb{E}[\xi_i] = 0$, and it satisfies $|\xi_i\ell_h(x_i, y_i)| = \ell_h(x_i, y_i) \le M$ by non-negativity of the loss and Proposition **??**, hence it satisfies the assumptions of Hoeffding's lemma. We obtain

$$\mathbb{E}\left[\exp\left(t\xi_i\ell_h(x_i,y_i)\right)\right] \le e^{t^2M^2/2}.$$

Because this holds for any $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, we have

$$\mathbb{E}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\,\middle|\,D\right] \le e^{t^2M^2/2}$$

surely. Plugging this into (2),

$$\exp\left(t\,\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) \le \mathbb{E}\left[\sum_{\ell_h\in\mathscr{L}}\prod_{i=1}^{n}e^{t^2M^2/2}\right] = |\mathscr{L}|\,e^{nt^2M^2/2}.$$

Taking the logarithm of both sides,

$$t\,\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right] \le \log|\mathscr{L}| + \frac{nt^2M^2}{2},$$

and dividing by $tn > 0$,

$$\mathcal{R}_n(\mathscr{L}) \le \frac{\log|\mathscr{L}|}{tn} + \frac{tM^2}{2} =: b(t) \quad (3)$$

Since (3) holds for every $t > 0$, we look for the tightest bound. We set the derivative in t to zero

$$\frac{\mathrm{d}b(t)}{\mathrm{d}t} = -\frac{\log|\mathscr{L}|}{t^2 n} + \frac{M^2}{2} = 0 \quad \Longrightarrow \quad t^\star = \sqrt{\frac{2\log|\mathscr{L}|}{nM^2}}.$$

Observe that since it is fair to assume $|\mathscr{L}| > 1$, the second derivative is nonnegative for any t

$$\frac{2\log|\mathscr{L}|}{t^3 n} \geq 0 \quad \forall t$$

hence $t^\star$ is a minimizer (if $|\mathscr{L}| = 1$, the bound follows by letting $t \to \infty$). Plugging $t^\star$ into (3) yields

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log|\mathscr{L}|}{n}}$$

□

Proof of Lemma **??**. Most of the proof is similar to Proof of Lemma **??**.

Let $(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n)) = \ell(D)$. For any $t > 0$,

$$\exp\left(t\,\mathbb{E}\left[\sup_{\ell(D)\in\mathscr{L}(D)} \sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i)\right]\right) \leq \mathbb{E}\left[\sum_{\ell(D)\in\mathscr{L}(D)} \prod_{i=1}^{n} \mathbb{E}\left[\exp\left(t\xi_i \ell_h(X_i, Y_i)\right) \,\middle|\, D\right]\right]$$

following the same steps as in Proof of Lemma **??**, all of which are independent of the data disorder, as they can be made fixing an observation S.

Now we apply Hoeffding's lemma (Theorem **??**). These steps also regard the argument of the expectation over the Rademacher variables disorder. We obtain

$$\mathbb{E}\left[\exp\left(t\xi_i \ell_h(X_i, Y_i)\right) \,\middle|\, D\right] \leq e^{t^2 M^2/2}.$$

surely as in Proof of Lemma **??**. Plugging this into (2),

$$\begin{aligned}
\exp\left(t\,\mathbb{E}\left[\sup_{\ell(D)\in\mathscr{L}(D)} \sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i)\right]\right) &\leq \mathbb{E}\left[\sum_{\ell(D)\in\mathscr{L}(D)} \prod_{i=1}^{n} e^{t^2 M^2/2}\right] \\
&= \mathbb{E}\left[N_D(\mathscr{H}) e^{nt^2 M^2/2}\right] \\
&= \mathbb{E}\left[N_D(\mathscr{H})\right] e^{nt^2 M^2/2}.
\end{aligned}$$

The rest of the proof follows exacltly as Proof of Lemma **??**, replacing $|\mathscr{L}|$ with $\mathbb{E}\left[N_D(\mathscr{H})\right]$. □

Proof of Lemma **??**. We prove that $N_S(\mathscr{H}) \le |\mathscr{H}|$ and $N_S(\mathscr{H}) \le 2^n$ for every $S \in (\mathcal{X} \times \mathcal{Y})^n$. Fix $S = (z_1, \dots, z_n)$ and consider the map

$$\Phi \colon \mathscr{H} \to \{0,1\}^n, \qquad h \mapsto \big(\ell_h(z_1), \dots, \ell_h(z_n)\big),$$

so that $N_S(\mathscr{H}) = |\Phi(\mathscr{H})|$. Since the image of a set under any map has cardinality at most that of the domain, $|\Phi(\mathscr{H})| \le |\mathscr{H}|$. Since $\Phi(\mathscr{H}) \subseteq \{0,1\}^n$, we also have $|\Phi(\mathscr{H})| \le |\{0,1\}^n| = 2^n$. Because this holds for every realization S of D, the claim for $N_D(\mathscr{H})$ follows. □

Proof of Theorem **??**. Since $\hat{h}_S \in \mathscr{H}$ for any realization S of D,

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le \mathbb{E}\left[\sup_{h \in \mathscr{H}} (R - \hat{R}_D)(h)\right]$$

The proof is concluded applying Theorem **??**, bounding the Rademacher complexity with Lemma **??**, and bounding $N_D(\mathscr{H})$ with Lemma **??**. □

Proof of Lemma **??**. Recall $\mathscr{L} = \{(x,y) \mapsto \ell(y, h(x)) : h \in \mathscr{H}\}$, where ℓ is the zero-one loss. Applying the first inequality of Theorem 5.1 of [2], with $\mathbb{S}_{\mathcal{F}}(X_1^{2n}) = N_{2n}(\mathscr{H})$, we obtain that with probability at least $1 - \delta$, simultaneously for all $h \in \mathscr{H}$,

$$R(h) - \widehat{R}_D(h) \le 2\sqrt{\frac{R(h)\log(4N_{2n}(\mathscr{H})/\delta)}{n}}.$$

Set

$$\delta_\varepsilon = 4N_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon < 1$, applying the preceding inequality with $\delta = \delta_\varepsilon$ gives

$$\Pr\left[\exists h \in \mathscr{H} : R(h) - \widehat{R}_D(h) > \varepsilon\sqrt{R(h)}\right] \le 4N_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon \ge 1$, this probability bound holds trivially. Finally, since $n\varepsilon^2 \ge 8$,

$$4e^{-n\varepsilon^2/4} \le 2e^{-n\varepsilon^2/16},$$

which proves the claim. □

Proof of Theorem **??**. Let $\delta \in (0,1)$ and set

$$u \coloneqq \frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n} \qquad \text{and} \qquad \varepsilon \coloneqq 4\sqrt{u},$$

so that $2N_{2n}(\mathscr{H})e^{-n\varepsilon^2/16} = \delta$. Since $N_{2n}(\mathscr{H}) \geq 1$ and $\delta < 1$, we have $n\varepsilon^2 = 16\log(2N_{2n}(\mathscr{H})/\delta) \geq 16\log 2 \geq 8$, hence Lemma **??** applies and gives

$$\Pr\left[\exists h \in \mathscr{H} : (R - \hat{R}_D)(h) > \varepsilon\sqrt{R(h)}\right] \leq \delta.$$

Therefore, with probability at least $1-\delta$, we have $(R - \hat{R}_D)(h) \leq \varepsilon\sqrt{R(h)}$ for all $h \in \mathscr{H}$, and in particular for $h = \hat{h}_D \in \mathscr{H}$.

On this event, we isolate $\hat{R}_D(\hat{h}_D)$. Let $r \coloneqq R(\hat{h}_D)$ and $\hat{r} \coloneqq \hat{R}_D(\hat{h}_D)$, so that $r - \varepsilon\sqrt{r} - \hat{r} \leq 0$. This is a quadratic inequality in $x \coloneqq \sqrt{r} \geq 0$, namely $x^2 - \varepsilon x - \hat{r} \leq 0$, which implies that x is at most the largest root,

$$\sqrt{r} \leq \frac{\varepsilon + \sqrt{\varepsilon^2 + 4\hat{r}}}{2}.$$

Squaring both sides,

$$r \leq \frac{2\varepsilon^2 + 4\hat{r} + 2\varepsilon\sqrt{\varepsilon^2 + 4\hat{r}}}{4} = \hat{r} + \frac{\varepsilon^2}{2} + \frac{\varepsilon}{2}\sqrt{\varepsilon^2 + 4\hat{r}} \leq \hat{r} + \varepsilon^2 + \varepsilon\sqrt{\hat{r}},$$

where in the last inequality we used $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$. Hence, substituting $\varepsilon = 4\sqrt{u}$, with probability at least $1-\delta$,

$$\begin{aligned}(R - \hat{R}_D)(\hat{h}_D) = r - \hat{r} &\leq \varepsilon\sqrt{\hat{r}} + \varepsilon^2 \\ &= 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2N_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

Proof of Corollary **??**. Recall $h_b = \arg\min_{h\in\mathscr{H}} R(h)$, $R(h_b) = R^*_{\mathscr{H}}$ by definition of $R^*_{\mathscr{H}}$ and $\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_b)$ by definition of the ERM $\hat{h}_D$. Since h_b is independent of the data, the variables $\ell(h_b(X_i), Y_i)$ are i.i.d. Bernoulli with mean $R^*_{\mathscr{H}}$ and variance $R^*_{\mathscr{H}}(1 - R^*_{\mathscr{H}}) \leq R^*_{\mathscr{H}}$. Bernstein inequality (Theorem **??**) with $M = 1$ gives, with probability at least $1 - \delta/2$,

$$\hat{R}_D(h_b) \leq R^*_{\mathscr{H}} + \sqrt{\frac{2R^*_{\mathscr{H}}\log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}.$$

Let $u \coloneqq \log(4N_{2n}(\mathscr{H})/\delta)/n$. Since $N_{2n}(\mathscr{H}) \geq 1$, $\log(2/\delta)/n \leq u$. Thus

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_b) \leq R^*_{\mathscr{H}} + 2\sqrt{R^*_{\mathscr{H}}u} + u = \left(\sqrt{R^*_{\mathscr{H}}} + \sqrt{u}\right)^2. \tag{4}$$

Theorem **??** at confidence $1 - \delta/2$ gives, with probability at least $1 - \delta/2$,

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\hat{R}_D(\hat{h}_D)\,u} + 16u. \tag{5}$$

By a union bound, both events hold simultaneously with probability at least $1-\delta$. Hence, plugging (4) in the RHS of (5), we obtain that with probability at least $1-\delta$,

$$\begin{aligned}(R-\hat{R}_D)(\hat{h}_D) &\le 4\sqrt{u}\left(\sqrt{R^*_{\mathscr{H}}}+\sqrt{u}\right)+16u \\ &= 4\sqrt{\frac{R^*_{\mathscr{H}}\log(4N_{2n}(\mathscr{H})/\delta)}{n}}+20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

Proof of Corollary **??**. Substitute $R^*_{\mathscr{H}}=\sigma^2+A_{\mathscr{H}}$ into Corollary **??**. □

Proof of Theorem **??**. Let $\text{gap}=(R-\hat{R}_D)(\hat{h}_D)$ and $C=\sqrt{\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}}$. Corollary **??** gives

$$\begin{aligned}&\Pr\left[\text{gap}\le 4\sqrt{R^*_{\mathscr{H}}}\,C+20C^2\right]\ge 1-\delta \\ \implies &\Pr\left[\text{gap}> 4\sqrt{R^*_{\mathscr{H}}}\,C+20C^2\right]\le \delta\end{aligned}$$

Set $\varepsilon=4\sqrt{R^*_{\mathscr{H}}}\,C+20C^2$. Let us now find $\delta(\varepsilon)$:

$$20C^2+4\sqrt{R^*_{\mathscr{H}}}\,C-\varepsilon=0 \implies C=\frac{-4\sqrt{R^*_{\mathscr{H}}}+\sqrt{16R^*_{\mathscr{H}}+80\varepsilon}}{40}=\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}$$

So we have

$$\begin{aligned}\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}&=\left(\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}\right)^2 \\ \frac{4N_{2n}(\mathscr{H})}{\delta}&=\exp\left(n\left(\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}\right)^2\right)\end{aligned}$$

and hence

$$\delta(\varepsilon)=4N_{2n}(\mathscr{H})\exp\left(-\frac{n}{100}\left(\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}\right)^2\right).$$

Using that $\mathbb{E}[X]=\int_0^\infty \Pr[X\ge\varepsilon]\,d\varepsilon$ for $X\ge 0$, applied to $\text{gap}^+\coloneqq\max\{\text{gap},0\}$,

$$\mathbb{E}[\text{gap}]\le\mathbb{E}[\text{gap}^+]=\int_0^\infty \Pr[\text{gap}\ge\varepsilon]\,d\varepsilon\le\int_0^\infty \min\{1,\delta(\varepsilon)\}\,d\varepsilon.$$

We now compute the integral by exchanging the roles of ε and δ. The map $\varepsilon\mapsto\delta(\varepsilon)$ is continuous and strictly decreasing on $[0,\infty)$, with $\delta(0)=4N_{2n}(\mathscr{H})>1$ and $\delta(\varepsilon)\to 0$ as $\varepsilon\to\infty$. Hence, for every $t\in(0,1)$ there is a unique $\varepsilon(t)>$

0 such that $\delta(\varepsilon(t)) = t$, and it is given by $\varepsilon(t) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ with $C = \sqrt{\log(4N_{2n}(\mathscr{H})/t)/n}$, by construction of $\delta(\varepsilon)$. Moreover, for $t \in (0,1)$,

$$\min\{1, \delta(\varepsilon)\} > t \iff \delta(\varepsilon) > t \iff \varepsilon < \varepsilon(t).$$

Writing $\min\{1, \delta(\varepsilon)\} = \int_0^1 \mathbf{1}\{t < \min\{1, \delta(\varepsilon)\}\}\, dt$ and exchanging the order of integration by Tonelli's theorem (the integrand is non-negative), we obtain

$$\int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon = \int_0^1 \int_0^\infty \mathbf{1}\{\varepsilon < \varepsilon(t)\}\, d\varepsilon\, dt = \int_0^1 \varepsilon(t)\, dt = \int_0^1 \varepsilon(\delta)\, d\delta.$$

Hence, writing $\varepsilon(\delta) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ explicitly,

$$\mathbb{E}[\text{gap}] \le \int_0^1 \varepsilon(\delta)\, d\delta = 4\sqrt{\frac{R^*_{\mathscr{H}}}{n}} \int_0^1 \sqrt{\log \frac{4N_{2n}(\mathscr{H})}{\delta}}\, d\delta + \frac{20}{n} \int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta.$$

The logarithm is the only part depending on δ, and its integral is

$$\int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta = \log(4N_{2n}(\mathscr{H})) - \int_0^1 \log \delta\, d\delta = \log(4N_{2n}(\mathscr{H})) + 1.$$

Applying Jensen's inequality to the square root term,

$$\int_0^1 \sqrt{\log \frac{4N_{2n}(\mathscr{H})}{\delta}}\, d\delta \le \sqrt{\int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta} = \sqrt{\log(4N_{2n}(\mathscr{H})) + 1},$$

we obtain

$$\mathbb{E}[\text{gap}] \le 4\sqrt{\frac{R^*_{\mathscr{H}}\left(\log(4N_{2n}(\mathscr{H})) + 1\right)}{n}} + 20\, \frac{\log(4N_{2n}(\mathscr{H})) + 1}{n}.$$

Asymptotically,

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\, \log N_{2n}(\mathscr{H})}{n}} + \frac{\log N_{2n}(\mathscr{H})}{n}\right)$$

because, at the numerator,

$$\log(4N_{2n}(\mathscr{H})) + 1 = \log 4 + \log N_{2n}(\mathscr{H}) + 1 = \log N_{2n}(\mathscr{H}) + O(1)$$

and for large x,

$$\log x + O(1) \le C \log x$$

for some constant $C > 0$. Apply Lemma **??**, obtaining

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le \min\left\{O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\log|\mathscr{H}|}{n}} + \frac{\log|\mathscr{H}|}{n}\right), O\left(\sqrt{A_{\mathscr{H}} + \sigma^2} + 1\right)\right\}$$

□

Proof of Lemma **??**. We prove the inequalities for $N_S(\mathscr{H})$, for every $S \in (\mathcal{X} \times \mathcal{Y})^n$. Since $N_D(\mathscr{H})$ takes the value $N_S(\mathscr{H})$ when D takes the value S, they hold for $N_D(\mathscr{H})$ for every realization of D.

Fix $S = ((x_1, y_1), \dots, (x_n, y_n)) \in (\mathcal{X} \times \mathcal{Y})^n$, and consider the maps

$$\Psi \colon \mathscr{H} \to \mathcal{Y}^n, \quad h \mapsto \big(h(x_1), \dots, h(x_n)\big),$$

$$\lambda \colon \mathcal{Y}^n \to \{0,1\}^n, \quad (\hat{y}_1, \dots, \hat{y}_n) \mapsto \big(\ell(\hat{y}_1, y_1), \dots, \ell(\hat{y}_n, y_n)\big).$$

The map $\Phi \colon h \mapsto (\ell_h(x_1, y_1), \dots, \ell_h(x_n, y_n))$ of the proof of Lemma **??** factorizes as $\Phi = \lambda \circ \Psi$, hence $N_S(\mathscr{H}) = |\Phi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))|$. Moreover, by definition of growth function, $|\Psi(\mathscr{H})| \leq \Pi_n(\mathscr{H})$.

$1 \leq N_S(\mathscr{H})$. Since $\mathscr{H}$ is nonempty, $\Phi(\mathscr{H})$ is nonempty.

$N_S(\mathscr{H}) \leq \Pi_n(\mathscr{H})$. Since the image of a set under any map has cardinality at most that of the set,

$$N_S(\mathscr{H}) = |\lambda(\Psi(\mathscr{H}))| \leq |\Psi(\mathscr{H})| \leq \Pi_n(\mathscr{H}).$$

$\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$. Let $x_1, \dots, x_n \in \mathcal{X}$ and let Ψ be the corresponding map. As above, $|\Psi(\mathscr{H})| \leq |\mathscr{H}|$, and $\Psi(\mathscr{H}) \subseteq \mathcal{Y}^n$ with $|\mathcal{Y}^n| = 2^n$, since $\mathcal{Y} = \{-1, 1\}$. Taking the supremum over $x_1, \dots, x_n$ gives $\Pi_n(\mathscr{H}) \leq |\mathscr{H}|$ and $\Pi_n(\mathscr{H}) \leq 2^n$.

$\Pi_n(\mathscr{H}) = N_n(\mathscr{H})$. Taking the supremum over S in $N_S(\mathscr{H}) \leq \Pi_n(\mathscr{H})$ gives $N_n(\mathscr{H}) \leq \Pi_n(\mathscr{H})$. For the converse, let $x_1, \dots, x_n \in \mathcal{X}$, take any $y_1, \dots, y_n \in \mathcal{Y}$, and let $S = ((x_1, y_1), \dots, (x_n, y_n))$. Since $\mathcal{Y} = \{-1, 1\}$ and ℓ is the zero-one loss, $\hat{y}_i = y_i$ if $\ell(\hat{y}_i, y_i) = 0$ and $\hat{y}_i = -y_i$ if $\ell(\hat{y}_i, y_i) = 1$. Hence λ is injective, and

$$|\Psi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))| = N_S(\mathscr{H}) \leq N_n(\mathscr{H}).$$

Taking the supremum over $x_1, \dots, x_n$ gives $\Pi_n(\mathscr{H}) \leq N_n(\mathscr{H})$. □

Appendix B. Preliminary experiments

B.1. **Noise structure.** We generate the labels with a centered parabola (see Section B.2), so that roughly half of them are 1 and half -1 in the interval $[0, 10]$. Then, we inspect what happens if we use a Bernoullian noise instead of a Gaussian one, i.e. when data is generated as

$$y \leftarrow sign(h^*(x))b$$

for $b \sim \text{Bernoulli}(p)$. Notice that this noise flips the labels with a probability of $1 - p$. While from the first experiments it seemed that the algorithm achieved a higher test error, we postulate that this may be due to the portion of misclassified inputs, rather than to the noise structure itself. Hence, fixing $p = 0.8$, we look for the value of σ that achieves roughly the same portion of mislcassified inputs (20%). Figure 1 shows that this happens for $\sigma = 5.5$, and Figures 2 and 3 show that the two noise structures then yield essentially the same train and test errors.

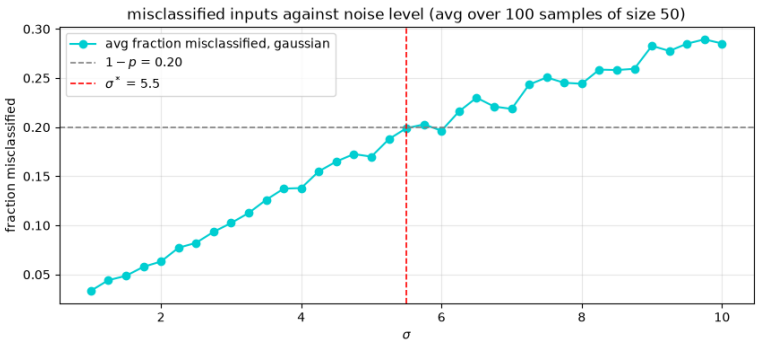


Figure 1. Calibrating Gaussian noise on the parabola. Average fraction of misclassified inputs against σ, over 100 samples of size $n = 50$.

B.2. **Data-generating function.** Our first choice of h^* is a parabola centered horizontally at 5, the midpoint of the support $\mathcal{X}$, and vertically so that roughly half of the points fall above the x-axis and half below. I.e.,

$$h^*(x) = (x - 5)^2 - 6.25$$

Figure 4 shows the data generated with the centered parabola, for different amounts of noise.

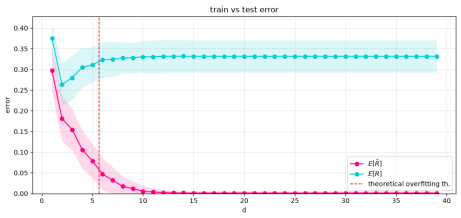


FIGURE 2. Parabola, Bernoulli noise with $p = 0.8$; $n = 50$, $N = 10,000$, $n_{\text{test}} = 200,000$.

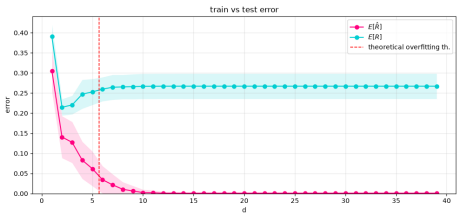


FIGURE 3. Parabola, Gaussian noise with $\sigma = 5.5$; $n = 50$, $N = 10,000$, $n_{\text{test}} = 200,000$.

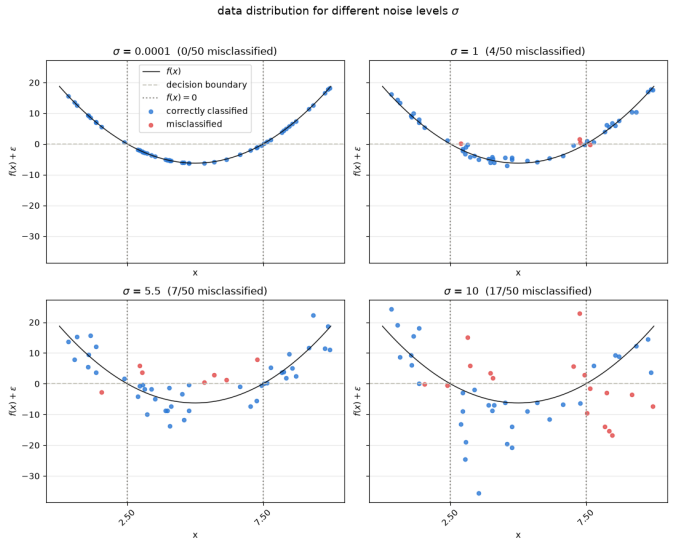


FIGURE 4. Samples from the parabola. One sample of size $n = 50$ for $\sigma \in \{10^{-4}, 1, 5.5, 10\}$.

Because in theory an algorithm could learn such a classification pattern as soon as it sees 3 points (one point for each region), we investigate what happens when the underlying function is more complex. We study a sinusoid with varying frequency,

$$h^*(x) = 5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right)$$

We multiply the sine by 5 to allow for enough height to properly inspect the data, while the argument of the sine is such that m is the number of times h^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of $1s$ and $-1s$.

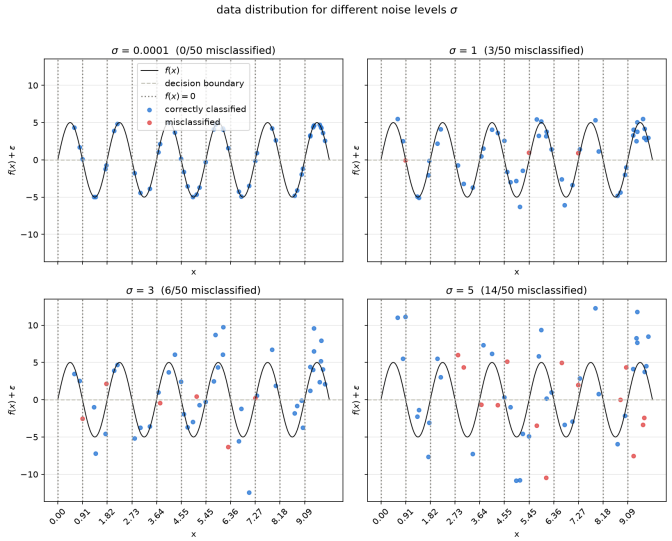


FIGURE 5. Samples from the sine ($m = 10$). One sample of size $n = 50$ for $\sigma \in \{10^{-4}, 1, 3, 5\}$.

Figure 5 shows the data generated with the sine for $m = 10$, for different amounts of noise.

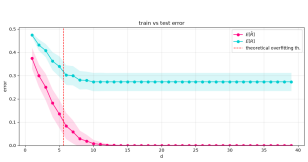


FIGURE 6. Sine, $m = 10$.

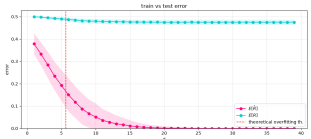


FIGURE 7. Sine, $m = 100$.

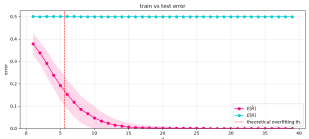


FIGURE 8. Sine, $m = 1000$.

We explore how a random sample and the performance of the algorithm change for different sine frequencies. In particular, we expect that for a high frequency the algorithm will not learn, as our datasize is $n = 50$. In Figures 6–8, we test a function that osciallets $10, 100$, and 1000 times with a noise level of $\sigma = 3$ (we reduce it compared to $\sigma = 5$, having increased the complexity of the function). We notice that for $m = 1000$ the algorithm fails to learn, and the test error stays stable at 0.5. This is because the labels change so frequently with respect to the number of points in the training dataset n that it is as if we had random labels. Hence, we

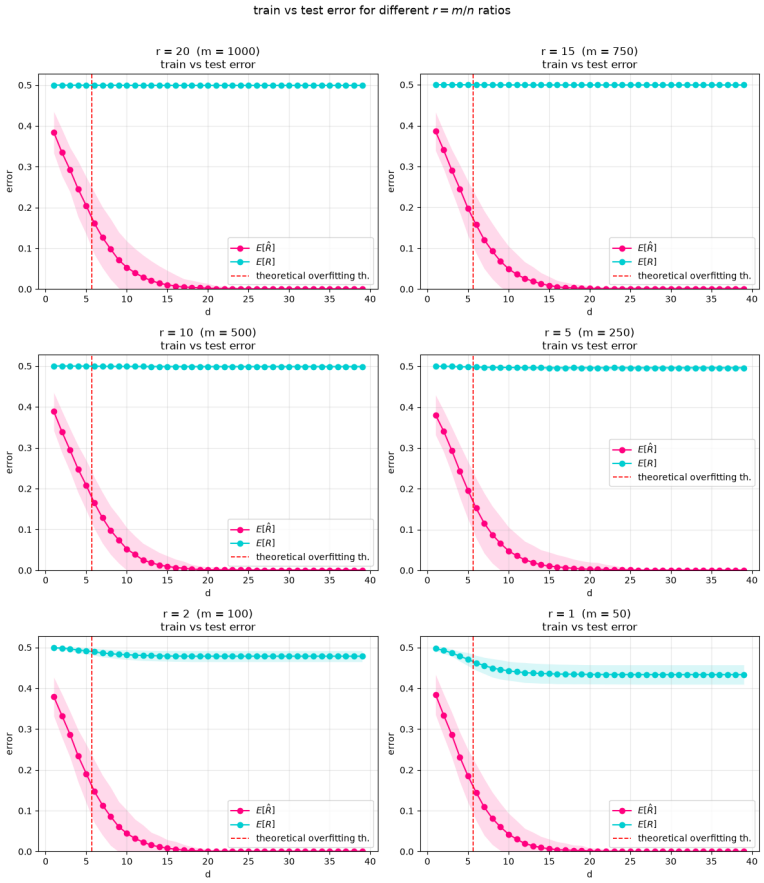


FIGURE 9. Learnability threshold in m/n. Train and test error for different ratios $r = m/n$; $n = 50$, $\sigma = 3$, $N = 10,000$, $n_{\text{test}} = 200,000$.

investigate the relation between the complexity of the underlying function m and the data size n. We find that for $\frac{m}{n} > 5$, roughly, the algorithm ceases to learn (Figure 9). In the following experiments, we fix $m = 10$ and $n = 50$, which is very far from the threshold.

We then verify whether a Gaussian noise with $\sigma = 5.5$ achieves the same portion of misclassified inputs on average as a Bernoullian noise with $p = 0.8$, as was the case for data generated with a parabola. Figure 10 shows that this is not the case on average (29% against 20%), while Figure 11 shows that the desired value is $\sigma = 3.5$.

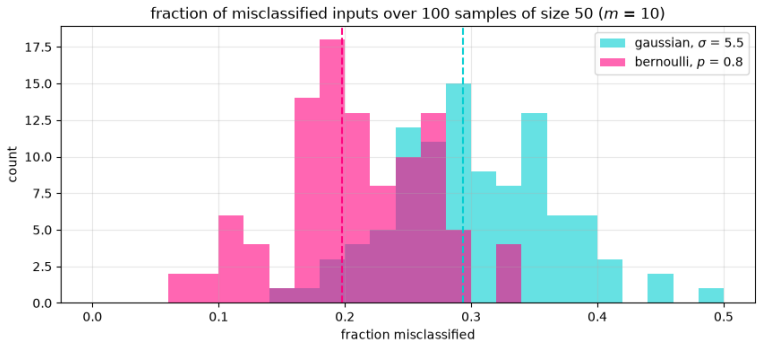


FIGURE 10. Gaussian against Bernoulli noise on the sine ($m = 10$). Fraction of misclassified inputs over 100 samples of size $n = 50$; dashed lines are the averages.

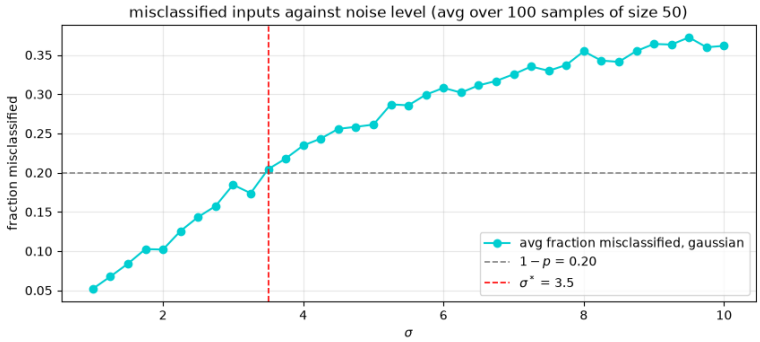


FIGURE 11. Calibrating Gaussian noise on the sine ($m = 10$). Average fraction of misclassified inputs against σ, over 100 samples of size $n = 50$.

Finally, we investigate the afore-mentioned intuition about how easy it is for the algorithm to learn a simple function. We take $m = 2$ (which corresponds to a parabola, in terms of classification pattern), sample uniformly from $\mathcal{X}$, and look for the training set size n for which the test error gets within 0.01 of the Bayes risk on average, at depth 2 and 3.

We start with little noise, $\sigma = 1$, for which the Bayes risk is ≈ 0.05 (Figure 12). Here the algorithm learns the function quickly: with $n = 50$ points the test error is already ≈ 0.07, within 0.02 of the Bayes risk, and it gets within 0.01 with $n = 200$ at depth 2 and $n = 500$ at depth 3.

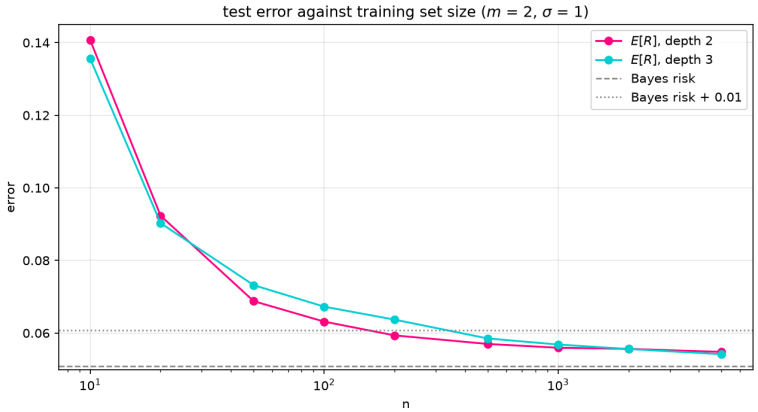


FIGURE 12. Sample size needed for a simple function, $\sigma = 1$. Test error at depth 2 and 3 against n for the sine with $m = 2$, averaged over 100 samples; $N = 10,000$, $n_{\text{test}} = 200,000$.

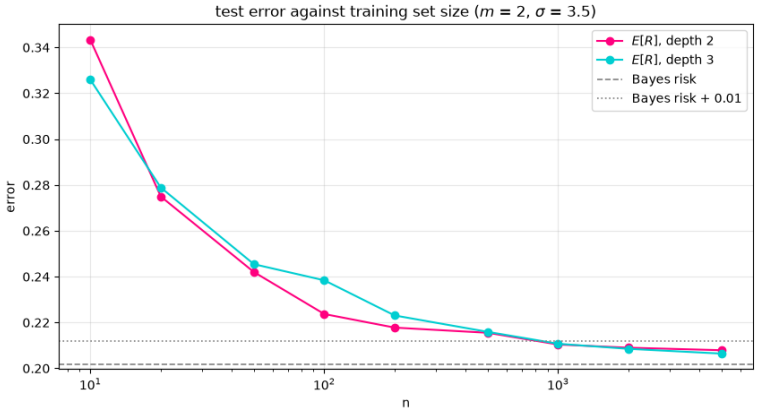


FIGURE 13. Sample size needed for a simple function, $\sigma = 3.5$. Test error at depth 2 and 3 against n for the sine with $m = 2$, averaged over 100 samples; $N = 10,000$, $n_{\text{test}} = 200,000$.

With our noise level, $\sigma = 3.5$, the Bayes risk is ≈ 0.20 and $n = 1000$ training points are needed on average for the test error to get within 0.01 of it, both at depth 2 and 3 (Figure 13). Hence, although two splits are enough to represent the function, with this amount of noise the algorithm needs many points to place them correctly.

Appendix C. Additional results

Fact C.1 (Bound on the Rademacher complexity of $\mathscr{L}$)**.** *Let ℓ take values in $[0, M]$ for some $M > 0$. Then, for any distribution p, any $\mathscr{H}$ and any $n \geq 1$,*

$$\mathcal{R}_n(\mathscr{L}) \leq \frac{M}{2}.$$

In particular, if ℓ is the zero-one loss, $\mathcal{R}_n(\mathscr{L}) \leq \frac{1}{2}$.

Proof. Fix a realization of D and ξ, and let $I_+ = \{i : \xi_i = 1\}$. For every $\ell_h \in \mathscr{L}$, since $0 \leq \ell_h(X_i, Y_i) \leq M$,

$$\frac{1}{n}\sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i) = \frac{1}{n}\sum_{i \in I_+} \ell_h(X_i, Y_i) - \frac{1}{n}\sum_{i \notin I_+} \ell_h(X_i, Y_i) \leq \frac{1}{n}\sum_{i \in I_+} \ell_h(X_i, Y_i) \leq \frac{M|I_+|}{n}.$$

The right-hand side does not depend on h, hence

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i) \leq \frac{M}{n}\sum_{i=1}^{n} \mathbf{1}(\xi_i = 1).$$

Taking expectations over ξ and D, and since $\Pr[\xi_i = 1] = \frac{1}{2}$,

$$\mathcal{R}_n(\mathscr{L}) \leq \frac{M}{n}\sum_{i=1}^{n} \Pr[\xi_i = 1] = \frac{M}{2}.$$

The zero-one loss takes values in $\{0, 1\}$, so the second claim follows with $M = 1$. □

Plot of Theorem ?? without fit. Figure 14 shows the plot with the noise-sensitive bound derived in Theorem **??**,

$$\min\left\{ 4\sqrt{\frac{(R^* + A_{\mathscr{H}})\,(\log(4|\mathscr{H}|) + 1)}{n}} + 20\,\frac{\log(4|\mathscr{H}|) + 1}{n}, \right.$$
$$\left. 4\sqrt{\frac{(R^* + A_{\mathscr{H}})\,((2n+2)\log 2 + 1)}{n}} + 20\,\frac{(2n+2)\log 2 + 1}{n} \right\},$$

for different noise levels and before fitting the constants in the function to our problem specification. As is evident, the curve is too high to provide a useful interpretation, reaching a value of 30.

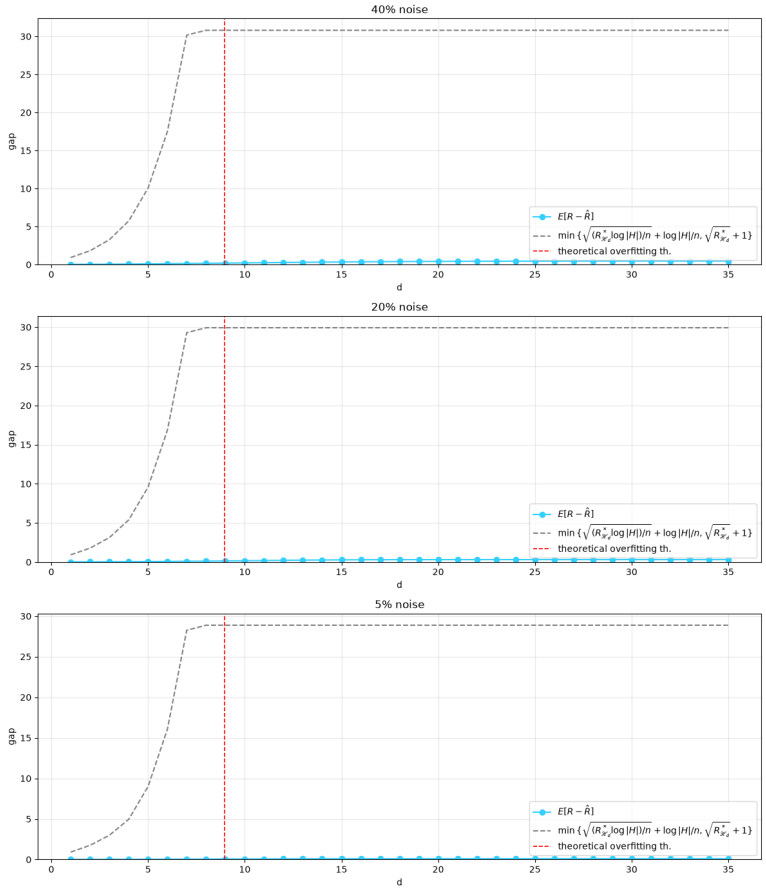


FIGURE 14. Noise and approximation error bound (Theorem **??**), before fitting constants

Appendix D. Code specifications

Computation of the best risk. The algorithm is explained using one-based indexing for the sake of exposition, while the real code uses zero-based indexing as usual in Python. We produce N noiseless labels for each input value in the support and we cluster the intervals over $\mathcal{X}$ where inputs have the same label. Let L be the number of contiguous blocks we obtain. We create a `dp` matrix with dimension $k_{\max} \times 2$: in the columns, we have the labels $y_j \in \{1, -1\}$ for $j = 1, 2$; in the rows, we have the number of leaves in the tree $k = 1, \dots, k_{\max}$, with $k_{\max} = \min(2^{d_{\max}}, L)$ the maximum number of leaves and $d_{\max}$ the largest depth we consider overall (that is a multiple of $\lceil \log_2 n \rceil$ as mentioned in Section **??**). Note that the tree reproducing exactly the noiseless labels has L leaves and has expected risk equal to $1 - p$, the portion of corrupted labels, reaching the Bayes risk. Since the best risk is bounded from below by the Bayes risk, considering more than L leaves cannot improve it. So, to avoid useless computations, we consider the minimum between L and $2^{d_{\max}}$. Given the true labels for each of the L blocks, it is easy to create also a `cost` matrix with dimension $L \times 2$, where `cost[i, j]` is the risk given by predicting y_j at block i. Let y_i be the true label in block i and ρ_i the portion of inputs falling into block i. We have

$$\texttt{cost[i,j]} = \begin{cases} \rho_i \cdot (1 - p) & \text{if } y_i = y_j \\ \rho_i \cdot p & \text{if } y_i \neq y_j \end{cases}$$

We initialize the `dp` matrix setting all entries to ∞. For the first iteration over block $i = 1$, we set

$$\texttt{dp[1, :]} = \texttt{cost[1, :]}$$

For each iteration $i = 2, \dots, L$ we dynamically update the `dp` matrix with minimum risks. Dynamically updating means that in `dp[k,j]` we store the cumulative minimum risk of having predicted label y_j at the *previous* block $i - 1$ with k leaves, and we create a new $k_{\max} \times 2$ matrix called `new` such that in `new[k,j]` we store the minimum risk of predicting label y_j at the current block i with k leaves. This minimum risk is computed as follows. Given that we predict y_j at block i, we choose between the tree that has predicted label y_j at block $i - 1$ using k leaves, i.e. that keeps the same label, and the tree that has predicted label $-y_j$ at block

$i-1$ using $k-1$ leaves, i.e. that switches labels. These two values are stored in the `dp` matrix, as said. Then, we add the cost of predicting label y_j in the current block i. This is summarized by the following line

$$\texttt{new[k,j]} = \min(\texttt{dp[k,j]}, \texttt{dp[k-1,3-j]}) + \texttt{cost[i,j]}$$

Note that $3-j$ swaps the indices 1 and 2, i.e. it stays for $-y_j$. Also, this code holds for $k > 1$, while for $k = 1$ the cost of switchiing is set to ∞, as it is not admissible. After passing over all $2k_{\max}$ entries of `new`, we update `dp = new` and pass to the next block $i+1$. The `dp` matrix will now contain the cumulative minimum risks for each possible label predicted in the last block i. Note that using an auxiliary matrix `new` is not strictly necessary, as we could update the `dp` matrix recursively, but using it eases understanding the logic. In the end, after passing over the last contiguous block, we obtain a matrix with the minimum risk for each number of leaves and each label assigned to the last block. We create a $k_{\max}$-long list `exact_k_risk` with the minimum value for each k,

$$\texttt{exact_k_risk[k]} = \min_{j \in \{1,2\}} \texttt{dp[k,j]}$$

Then, we create another $k_{\max}$-long list `at_most_k_risk` that stores the cumulative minimum for each possible *maximum* number of leaves,

$$\texttt{at_most_k_risk[k]} = \min_{r \in [1,k]} \texttt{exact_k_risk[r]}$$

Thus, the k-th entry of `at_most_k_risk` contains the best risk allowing at most k leaves. Finally, for each depth d we set $k_d = \min(2^d, L)$ and select the k_d-th entry in `at_most_k_risk`, yielding the best risk among trees with maximum depth d.